



JOBSHEET 2

Variables, Data Types, Operators (Arithmetic) and Input-Output

1. Learning Outcomes

- Students understand and explain Data Types in Java programming
- Students understand and explain variables in Java programming
- Students can explain input-output in Java programming
- Students can express and explain Sequences in Java programming
- Students understand and explain Arithmetic Operators in Java programming

2. Labs Activities

2.1 Experiment 1: Use of Variables

Allocated time: 40 minutes

1. Open the **text editor** or Visual Studio Code
2. Create a new file, name it **ExampleVariabelIDNumber.java**
3. Write the basic structure of the Java programming language which contains the **main()** function
4. Write the code below in **public static void main (String args [])**

```
String oneOfMyHobbies = "Playing futsal";
boolean isSmart = true;
char gender = 'M';
byte _age = 20;
double $gpa = 3.38, height = 1.68;
System.out.println(oneOfMyHobbies);
System.out.println("Are you smart ?" + isSmart);
System.out.println("Gender: "+gender);
System.out.println("My current age is "+ _age);
System.out.println(String.format("My GPA is %s and my height is %s meters", $gpa, height));
```

5. Run the program code that you have written, then observe the results

Questions!

1. Change the variable name so that the variable naming model is good and correct!
2. What is the function of %s in the statement below?

```
System.out.println(String.format("My GPA is %s and my height is %s meters", $gpa, height));
```

Is there anything you can use instead of %s? Name and explain it!

2.2 Experiment 2: Use of Data Types

Experiment time: 40 minutes

1. Open the Visual Studio Code
2. Create a new file, name it **ExampleDataTypeIDNumber.java**
3. Write the basic structure of the Java programming language which contains the **main()** function
4. Write the code below in **public static void main (String args [])**

```
char bloodGroup = 'A';
byte distance = (byte) 130;
short theNumberOfResidents = 1025;
float temperature = 60.50F;
double weight = 0.5467812345;
long balance = 150000000;
int number = 0x10;
System.out.println("Blood group\t\t: " + (byte) bloodGroup);
System.out.println("Distance\t\t: " + distance);
System.out.println("The number of residents\t: " + theNumberOfResidents);
System.out.println("Temperature\t\t: " + temperature);
System.out.println("Weight\t\t\t\t: " + (float) weight);
System.out.println("Balance\t\t\t\t: " + balance);
System.out.println("Number\t\t\t\t: " + number);
```

5. Run the program code that you have written, then observe the results

Questions!

1. Explain why the **bloodGroup** does not display an "A"!
2. Explain the meaning of **byte distance = (byte) 130**! Then, explain why the results change when displayed!
3. In the syntax **float temperature = 60.50F**; remove the letter **F**, then run again. What happened?
4. Why does the result change when displaying weight values?
5. Explain the meaning of initializing **0x10** on **number** variables! What does it do?

2.3 Experiment 3: Use of Operators

Experiment time: 40 minutes

1. Open the Visual Studio Code
2. Create a new file, name it **ExampleOperatorIDNumber.java**

3. Write the basic structure of the Java programming language which contains the **main()** function
4. Write the code below in the **public static void main (String args [])**

```
int a = 60;
int b = 20;
int c = 10;

System.out.println("Arithmetic Operators Example");
System.out.println("Value of a = " + a);
System.out.println("Value of b = " + b);
System.out.println("Value of c = " + c);

System.out.println("a + b + c = " + (a + b + c));
System.out.println("a - b - c = " + (a - b - c));
System.out.println("a * c      = " + (a * c));
System.out.println("a / b      = " + (a / b));
System.out.println("a % b      = " + (a % b));
System.out.println("a + b * c = " + (a + b * c));
System.out.println("(a + b) * c = " + ((a + b) * c));
```

5. Run the program code that you have written, then observe the results

Questions!

1. Explain, in your opinion, the difference between $a + b * c$ and $(a + b) * c$.
2. What is the difference between a / b and $a \% b$?

2.4 Experiment 4: Case Study

Experiment time: 30 minutes

Pay attention to the Case Study below!

Mr. Dani has a garage with a triangular shape. Mr. Dani plans to cement the ground floor of the garage so that it can be used to park motorbikes comfortably. Identify the input, output, and process to help Mr. Dani calculate the area of his garage, then implement it into the program code!

1. Identify input, output, and process

Input: base, height

Output: area

Process:

- a. Input base, height

- b. Calculate area = $1/2 * \text{base} * \text{height}$
- c. Output area

2. Identify the variables and data types used

Variables	Data Types
base	int
height	int
area	float

3. Implementation into program code

Program code:

1. Create a new file, name it **TriangleIDNumber.java**
2. Write the basic structure of the Java programming language which contains the **main()** function
3. Add the Scanner library at the top (outside) of the class

```
import java.util.Scanner;
```

4. Create a Scanner declaration inside the **main()** function

```
Scanner sc = new Scanner(System.in);
```

5. Create **int** variables for the **base** and **height**, then a **float** variable for the **area**.

```
int base, height;  
float area;
```

6. Write the command to input the base and height

```
System.out.print(s:"Input base: ");  
base=sc.nextInt();  
System.out.print(s:"Input height: ");  
height = sc.nextInt();
```

7. Write down the command to calculate the area of a triangle

```
area = base * height / 2;
```

8. Display the contents of the **area** variable

```
System.out.println("Area of the triangle: " + area);
```

9. Run the program code that you have written, then observe the results

Questions!

1. Explain why you have to declare Scanner in Experiment 4?
2. Explain the use of the program snippets below!

```
base=sc.nextInt();
```

```
height = sc.nextInt();
```

2.5 Experiment 5: Case Study

Experiment time: 30 minutes

Pay attention to the Case Study below!

Mrs. Dina is one of ABC bank customers who saved Rp. 5 million. The bank provides interest of 2% every year. Mrs. Dina saved for 5 years. How much interest and savings can you take now?

1. Identify input, output, and process

Input: base, height

Output: area

Process:

Input: initial savings amount, savings period

Output: interest, final savings amount

Other data: interest percentage = 0.02

Process:

- a. Input the initial savings amount, savings period
- b. Calculate interest = savings period * interest percentage * initial savings amount
- c. Calculate the final savings amount = interest + initial savings amount
- d. Interest output and final savings amount

2. Identify variables and data types

Variables	Data types
init_sav_amount	int
sav_period	int
final_sav_amount	double
interest	double
interest_percent = 0.02	double

3. Implementation into program code

Program code:

1. Create a new file, name it **BankIDNumber.java**
2. Write the basic structure of the Java programming language which contains the **main()** function
3. Add the Scanner library at the top (outside) of the class

```
import java.util.Scanner;
```

4. Create a Scanner declaration inside the **main()** function

```
Scanner sc = new Scanner(System.in);
```

5. Create **int** variables for the **base** and **height**, then a **float** variable for the **area**.
6. Create **int** variables for the **init_sav_amount** and **sav_period**, then **double** variables for the **final_sav_amount**, **interest**, and **interest_percent** according to the identification of variables and data types that has been done previously.

```
int init_sav_amount, sav_period;
double final_sav_amount, interest, interest_percent = 0.02;
```

7. Write the command to input the **init_sav_amount** and **sav_period**

```
System.out.print(s:"Input your initial savings amount: ");
init_sav_amount = sc.nextInt();
System.out.print(s:"Input your savings period: ");
sav_period = sc.nextInt();
```

8. Write down the command to calculate **interest**

```
interest = sav_period * interest_percent * init_sav_amount;
```

9. Write a command to calculate the **final_sav_amount**

```
final_sav_amount = interest + init_sav_amount;
```

10. Display the value of the **interest** and **final_sav_amount** variables

```
System.out.println("Interest: "+ interest);
System.out.println("Final savings amount: "+ final_sav_amount);
```

11. Run the program code that you have written, then observe the results



3. Assignment

1. Case Study 1:

Mrs. Jesi is an employee at PT. ABCD with a basic salary of Rp. 3,000,000 per month. At PT. ABCD, every employee receives a child allowance based on the number of children they have. The allowance is Rp. 150,000 per child per month. In addition, employees are required to contribute 5% of their basic salary each month to a mandatory pension fund.

Task: Write a program to calculate Mrs. Jesi's net monthly salary if she has 3 children.

2. Modify the program from Case Study 1 by making the basic salary, child allowance, and number of children dynamic inputs.

3. Case Study 2:

Mr. Jaka owns a piece of land with a width of 50 meters and a length of 100 meters. He plans to build two fish ponds: one circular pond with a diameter of 2 meters and one square pond with a side length of 2 meters. The remaining land will be planted with garden grass.

Task: Write a program to calculate the total area of land that will be planted with grass.

4. Modify the program from Case Study 2 by making the width, length, diameter, and side length dynamic inputs.